

AI Acceptance in the Sharing Economy

Data, Methodology and Progress against the Brief — Working Report

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1 The idea

Peer-to-peer rental marketplaces — homes (Airbnb, Vrbo), cars and RVs (Turo, Getaround, RVshare, Outdoorsy), and creative gear (Fat Llama, KitSplit, Lensrentals) — run entirely on **trust between strangers**, and they are now inserting **AI and AI agents** into that relationship: automated guest screening, chatbots, dynamic pricing, automated messaging. Our question:

How do users perceive AI agents in peer-to-peer rental platforms, and what drives or erodes their trust — across rental types, and across open community discussion (Reddit) versus paying-customer reviews (Trustpilot)?

We collect two complementary corpora. **Reddit is the baseline** — where users deliberate, complain and speculate in their own words. **Trustpilot is the verification layer** — structured reviews carrying a **1–5 star rating that acts as a ground-truth sentiment label**. Where the open discussion and the labelled reviews agree we gain confidence; where they diverge we learn something about the gap between discourse and experience.

2 Data source 1 — Reddit (baseline)

Collected with **RedditExtractor** (public Reddit API). The corpus went through three passes, each fixing the previous one:

Table 1: Reddit passes. Files: reddit_all.csv, reddit_targeted.csv, reddit_clean.csv.

Pass	Script	Rows	Status
v1 keyword (global)	scraping_data_v1.R	14,090	baseline only (off-topic)
v2 subreddit-targeted	01_scrape_reddit.R	1,009	raw, on-topic
v2 cleaned + themed	05_clean_reddit.R	314	USED for analysis

Why v1 failed. A single global keyword search ("AI rental platform helpful automation trust") returned ~14,090 mostly off-topic comments (K-pop, gaming, politics); under 1% mentioned “rental”. Kept only as a transparency baseline.

How we targeted (v2). We search *inside* topic-relevant subreddits instead of all of Reddit, using **47 (subreddit, keyword) pairs** across six layers:

- **Home / stay** — r/AirBnB, r/airbnb_hosts, r/AirBnBHosts, r/vrbo, r/ShortTermRentals

- **Car / RV** — r/Turo, r/gigwork, r/RVing, r/GoRVing
- **Camera / film / drone** — r/photography, r/AskPhotography, r/cinematography, r/videography, r/Filmmakers, r/drones, r/dronephotography, r/DJI
- **Event tech** — r/eventproduction, r/EventProduction, r/livesound
- **AI in commerce** — r/Entrepreneur, r/SaaS, r/startups, r/automation, r/ChatGPT, r/OpenAI, r/artificial, r/singularity
- **Trust / privacy** — r/privacy, r/technology

Keywords centred on AI, automation, hosting, rental and trust (e.g. “AI”, “host AI”, “automated message”, “rental AI”, “AI agent”). Parameters: 30 threads per search, `period = "all"`, threads kept only if `comments >= 6`, capped at 400 threads; deleted/removed and sub-20-character comments dropped. The keyword that *finds* a thread is never used as its label — sentiment is derived from the text downstream.

How we balanced and cleaned. Two viral subreddits (r/Filmmakers, r/Entrepreneur) hosted threads about AI replacing humans in creative/business careers — off-topic for rental trust — so `05_clean_reddit.R` drops them and tags each remaining comment `sharing_economy` or `ai_tech`. (The scrape can also cap comments per thread at 120 to stop any single thread dominating.)

The dataset (`reddit_clean.csv`): 314 comments, 17 subreddits, 38 threads, Oct 2012 – May 2026, mean length 239 characters. Themes are close to balanced (below); the top subreddits are r/ChatGPT (81), r/photography (76), r/privacy (56), r/GoRVing (38), r/AirBnB (22), r/dji (13).

Table 2: Reddit theme balance.

Theme	Comments	Share
ai_tech	146	46%
sharing_economy	168	54%

3 Data source 2 — Trustpilot (verification)

Trustpilot gives what Reddit cannot: a **1–5 star rating on every review** — a ground-truth label we did not have to infer. We can validate a sentiment lexicon against these stars, then carry the validated lexicon to the unlabelled Reddit comments (convergent validity).

How we scraped it. Trustpilot blocks naive scraping, so we used the Apify actor `casper11515/trustpilot-reviews-scraper` via the **Apify web UI, once per company**, downloading each result CSV into `data/trustpilot_raw/`. `02_merge_trustpilot.R` then merges them (robust to BOM/UTF-8 export issues), normalises columns, and keeps English reviews with real text and unique IDs -> `data/trustpilot_reviews.csv`.

The dataset: 3,912 unique English reviews across 7 platforms, spanning Jul 2019 – May 2026. Each row carries a star rating, title, body, reviewer country and (where present) the company reply.

Table 3: Trustpilot reviews per platform, with mean star rating.

Platform	Reviews	Mean star
Turo	1482	4.62
Fat Llama	835	4.11
RVshare	773	3.98
Outdoorsy	751	3.68

Platform	Reviews	Mean star
Lensrentals	35	2.94
Getaround	31	4.35
KitSplit	5	1.00

Ratings collapse to **706 negative** (1-2 stars), **139 neutral** (3), **3067 positive** (4-5). The strong positive skew is typical of opt-in review sites — we down-sample the positive class for balanced classification.

4 Progress against the assignment brief

Each task below is tagged **Done**, **Partly done**, **In progress** or **To do**. Our two classes are the Reddit theme (`sharing_economy` vs `ai_tech`) and, on Trustpilot, positive vs negative reviews (the star rating gives the label). Owners are still to be assigned by the team.

4.1 Dataset and preprocessing

Selection — Done. Two self-scraped corpora, no X/Twitter content: a targeted Reddit discussion corpus (`reddit_clean.csv`, 314 themed comments) and 3,912 Trustpilot reviews with 1–5 star labels across seven rental platforms.

Cleaning — Partly done; full pass next. Collection already strips deleted/removed bodies, sub-20-character comments and duplicate IDs. The normalisation the brief specifies — lowercasing, removing punctuation, numbers and stop words, tokenising, and stemming / lemmatising — is the immediate next script (`06_preprocess.R`), emitting unigram and bigram token tables for both corpora.

4.2 Exploratory analysis

Word frequency — Partly done. A first top-terms bar chart exists (`figures/fig1_topwords.png`, split by source). The brief’s exact form — top 10 words per class as side-by-side bars — will be regenerated from the cleaned tokens once preprocessing lands.

Word clouds — Partly done. A comparison cloud (words coloured by the source where they are relatively more frequent) is already produced; still to add are the commonality cloud (shared words) and the bigram variants, per class.

4.3 Structural and sentiment analysis

Word networks — To do. Per-class co-occurrence networks via `widyr::pairwise_count` + `ggraph`, showing how key terms cluster spatially.

Sentiment — Partly done. A Trustpilot rating-over-time timeline exists. The brief’s single-graph sentiment comparison of both classes (AFINN and NRC) is still to do; uniquely, we then validate that lexicon against Trustpilot’s star labels before trusting it on the unlabelled Reddit text.

4.4 Advanced modeling

Topic modelling (LDA) — To do. 3–5 topics per class on unigrams and bigrams, with trust-related seed words and a `text2vec` cross-check; the small Reddit corpus (38 threads) realistically caps us nearer 3–4 topics there.

Text embeddings — To do. GloVe / Word2Vec via `text2vec` to surface semantic clusters and the nearest neighbours of *trust*, *AI* and *host* inside the identified topics.

4.5 Synthesis and reporting

Integration — To do. Weave networks, sentiment and topics into one narrative. The spine is the cross-source triangulation: validate the lexicon on Trustpilot stars, apply it to Reddit, then compare which topics drive negativity in each source — agreement across two structurally different corpora is stronger evidence than either alone.

Strategic insights — To do. Turn the class differences into actionable recommendations for AI-agent design on rental platforms — a short set of concrete “design cards” on where automation builds versus erodes trust.

Literature — Partly identified. Trust and adoption frameworks are already in scope (Mayer et al. 1995 — ability / benevolence / integrity; Davis’s TAM; UTAUT) and will be cited alongside text-mining method references in the final report.

Report — In progress. This document covers data and methods; the final report expands the exploratory, modelling, synthesis and literature items above into full results.

5 Limitations

- Trustpilot skews positive (opt-in bias) — we down-sample for classification.
- Equipment-rental platforms (Lensrentals, KitSplit) have few reviews -> weak per-platform claims there.
- One large r/ChatGPT thread pulls AI-side topics toward “AI tools”; few Reddit threads overall (38) cap LDA at 3–4 topics per theme.
- “Trust” is mostly *implicit* (host/renter, payment, review language), measured via topics and sentiment, not keyword counts. English-only.
- Reddit’s anti-bot wall blocks fresh anonymous re-scrapes (needs an OAuth app; see `legacy/`). Turo/Getaround per-platform review dates need a parse check in `02_merge_trustpilot.R` before any per-platform time-series.

Repository: <https://github.com/tyomachkaa/ai-acceptance-sharing-economy> · Online Content Analysis, WU Vienna, 2026.